Optimal Reconfiguration of Systems in Aeronautics

Lecture 08: "Matrix-Sign-Function for the Algebraic Riccati Equation"

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Outline

1. Matrix Sign Function
2. B747 Application Example

Cuprins

1. Matrix Sign Function
   - Preliminaries on Matrix Functions
   - Matrix-Sign-Function Approach
2. B747 Application Example

Objective I

This lecture provides some results on Matrix-Sign-Function.

It is useful when the Matrix-Sign-Function approach is used for the implementation of the Progressive Accommodation Scheme.

It gives the properties of matrix functions, of Matrix-Sign-Function (MSF), and of spectral-projectors, respectively, while it also addresses the connections between the MSF and the Continuous Algebraic Riccati Equation respectively the MSF and spectral-projection, as well.

Properties of Matrix Functions I

Given matrix $A_f \in \mathbb{R}^{n\times n}$, the matrix-function $f(A_f) \in \mathbb{R}^{n\times n}$ (assume it is defined on the spectrum of A_f, $\Lambda(A_f)$; of course, $f(A_f) \in \mathbb{R}^{q\times q}$ or $f(A_f) \in \mathbb{R}^{q}$, with $q \neq n$, is possible depending on the definition of f) shows the following properties:

(i) $f(A_f)$ commutes with A_f, respectively

$$f(A_f)A_f = A_f f(A_f)\,;$$

(ii)

$$f\left(A_f^\top\right) = [f(A_f)]^\top\ ;$$

(iii) For any nonsingular T_f, it does hold that

$$f\left(T_f A_f T_f^{-1}\right) = T_f f(A_f)\, T_f^{-1}\,;$$

Properties of Matrix Functions II

iv.

$$f\left[\text{diag}\left(A_{f_1}, A_{f_2}, \ldots, A_{f_{\bar{n}}}\right)\right] = \text{diag}\left[f(A_{f_1}), f(A_{f_2}), \ldots, f(A_{f_{\bar{n}}})\right] ;$$

v. If A_f is diagonalizable, with

$$T_f^{-1} A_f T_f = D_f = \text{diag}\left(d_{f_1}, d_{f_2}, \ldots, d_{f_n}\right),$$

then

$$f(A_f) = T_f f(D_f) T_f^{-1} = T_f \text{diag}\left[f(d_{f_1}), f(d_{f_2}), \ldots, f(d_{f_n})\right] T_f^{-1} .$$

Properties of Matrix-Sign-Function I

Given matrix $A_f \in \mathbb{R}^{n \times n}$, the matrix-sign-function $Z_f \triangleq \text{sign}\,(A_f)$ has the following properties:

i. $Z_f^2 = I_n$ (Z_f is involutory);

ii. Z_f is diagonalizable, with eigenvalues ± 1;

iii. $Z_f A_f = A_f Z_f$;

iv. If A_f is real, then Z_f is real;

v. If Z_f is symmetric positive-definite, then $\text{sign}\,(A_f) = I_n$.

Properties of Spectral-Projectors I

Definition (Matrix-Projector)

Matrix $P_f \in \mathbb{R}^{n\times n}$ is called projector onto the subspace $\mathcal{S}_f \subset \mathbb{R}^{n\times n}$ if

$$\mathrm{Im}\,(P_f) = \mathcal{S}_f \text{ and } P_f^2 = P_f. \tag{1}$$

□

Definition (Spectral-Projector)

Let $H_f \in \mathbb{R}^{n\times n}$, with $\Lambda(H_f) = \Lambda_f^- \cup \Lambda_f^+$, $\Lambda_f^- \cap \Lambda_f^+ = \emptyset$, and let $\mathcal{S}_f^-$ be the (right) H_f-invariant-subspace corresponding to Λ_f^-.

Then a projector onto $\mathcal{S}_f^-$ is called a spectral-projector. □

Properties of Spectral-Projectors II

Lemma (Lancaster and Rodman; 1991)

Let $H_f \in \mathbb{R}^{n\times n}$ and let $P_f \in \mathbb{R}^{n\times n}$ be a spectral-projector onto the right H_f-invariant-subspace corresponding to Λ_f^-.

Then:

i. $Im\,(P_f) = Im\,(H_f P_f)$;
ii. a. $Ker\,(P_f) = Im\,(I_n - P_f)$;
    b. $Im\,(P_f) = Ker\,(I_n - P_f)$;
iii. $I_n - P_f$ *is a spectral-projector onto the right H_f-invariant-subspace corresponding to Λ_f^+.*

□

Matrix-Sign-Function and Spectral-Projection I

Computations based on the matrix-sign-function can be considered as spectral-projection methods as they involve

$$P_f^- = \frac{1}{2}\left[I_{2n} - \text{sign}\left(H_f\right)\right],$$

which is a spectral-projector onto the stable H_f-invariant-subspace.

Also

$$P_f^+ = \frac{1}{2}\left[I_{2n} + \text{sign}\left(H_f\right)\right]$$

is a spectral-projector onto the H_f-invariant-subspace corresponding to the eigenvalues in the open right-half plane.

Algebraic Riccati Equation and Matrix-Sign-Function I

Theorem (Byers; 1987)

The solution X_f *to*

$$A_f^\top X_f + X_f A_f - X_f B_f R_n^{-1} B_f^\top X_f + Q_n = 0 \tag{2}$$

results by solving the system

$$\begin{bmatrix} & Z_{f_{12}} \\ Z_{f_{22}} & + I_n \end{bmatrix} X_f = - \begin{bmatrix} Z_{f_{11}} & + I_n \\ Z_{f_{21}} & \end{bmatrix}, \tag{3}$$

where

$$sign\,(H_f) \triangleq Z_f = \begin{bmatrix} Z_{f_{11}} & Z_{f_{12}} \\ Z_{f_{21}} & Z_{f_{22}} \end{bmatrix},$$

for

$$H_f = \begin{bmatrix} A_f & -G_f \\ -Q_n & -A_f^\top \end{bmatrix}, \text{ with } G_f = B_f R_n^{-1} B_f^\top .$$

□

Matrix-Sign-Function and Jordan decomposition I

For the next development, introduce first the classical theorem on Jordan decomposition and the definition of a matrix function based on its Jordan decomposition.

Matrix-Sign-Function and Jordan decomposition II

Theorem (Jordan Decomposition)

If matrix $A_f \in \mathbb{R}^{n\times n}$ has $\bar{n}_f$ linearly independent eigenvectors, then it is similar to a matrix which is in Jordan form with $\bar{n}_f$ square blocks on the diagonal, respectively

$$J_f = T_f^{-1} A_f T_f = \begin{bmatrix} J_{f_1} & & \\ & \ddots & \\ & & J_{f_{\bar{n}_f}} \end{bmatrix}. \tag{4}$$

Each block has one eigenvector, one eigenvalue, and ones just above the diagonal, respectively

$$J_{f_k}(\lambda_{f_k}) = \begin{bmatrix} \lambda_{f_k} & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_{f_k} \end{bmatrix}. \tag{5}$$

When the block $J_{f_k}(\lambda_{f_k})$ has order $m_{f_k} > 1$, the eigenvalue λ_{f_k} is repeated m_{f_k} times and there are $m_{f_k} - 1$ of ones above the diagonal, with $m_{f_1} + m_{f_2} + \ldots + m_{f_{\bar{n}_f}} = n$; the same eigenvalue λ_{f_k} may appear in several blocks if it corresponds to several independent eigenvectors. $\square$

Matrix-Sign-Function approach I

Definition (Jordan Canonical Form and Matrix Function)

Let $A_f \in \mathbb{R}^{n\times n}$ have the Jordan canonical form (4) with Jordan blocks as in (5), where T_f is nonsingular.

Then

$$f(A_f) = T_f f(J_f)\, T_f^{-1} = T_f \operatorname{diag}\left\{ f\left[J_{f_k}(\lambda_{f_k})\right]\right\} T_f^{-1},$$

where

$$f\left[J_{f_k}(\lambda_{f_k})\right] = \begin{bmatrix} f(\lambda_{f_k}) & f'(\lambda_{f_k}) & \dots & \frac{f^{\left(m_{f_k}-1\right)}(\lambda_{f_k})}{\left(m_{f_k}-1\right)!} \\ & \ddots & \ddots & \vdots \\ & & \ddots & f'(\lambda_{f_k}) \\ & & & f(\lambda_{f_k}) \end{bmatrix};$$

the prime superscript $(.)'$ stands for derivation and $(.)!$ denotes the factorial product. $\square$

Matrix-Sign-Function approach II

Next, define the sign-function of a matrix based on its Jordan decomposition.

Definition (Matrix Sign Function)

If $A_f = T_f J_f T_f^{-1} \in \mathbb{R}^{n\times n}$ is a Jordan canonical form (4) so that

$$J_f = \begin{bmatrix} J_f^- & O_{k,n-k} \\ O_{n-k,k} & J_f^+ \end{bmatrix},$$

where block-matrices $J_f^- \in \mathbb{R}^{k\times k}$ and $J_f^+ \in \mathbb{R}^{(n-k)\times(n-k)}$ have spectra $\Lambda(J_f^-) \subset \mathbb{C}^-$ and $\Lambda(J_f^+) \subset \mathbb{C}^+$, then the sign-function of A_f writes as

$$\text{sign}\,(A_f) = T_f \begin{bmatrix} -I_k & O_{k,n-k} \\ O_{n-k,k} & I_{n-k} \end{bmatrix} T_f^{-1}.$$

If A_f has any pure imaginary eigenvalues, then $\text{sign}\,(A_f)$ is not defined. $\square$

Matrix-Sign-Function approach III

In the following, let the real $n \times n$ matrix $Z_f = \text{sign}\,(A_f)$ be defined and see that $Z_f^2 = I_n$ (Z_f is involutory); moreover, Z_f is diagonalizable with eigenvalues ± 1.

Also, reconsider the Hamiltonian-matrix (6), respectively

$$H_f = \begin{bmatrix} A_f & -G_f \\ -Q_n & -A_f^\top \end{bmatrix}, \tag{6}$$

where

$$G_f = B_f R_n^{-1} B_f^\top ,$$

and note that matrix X_f is the desired solution to the post-fault CARE if and only if

$$\begin{bmatrix} A_f & -G_f \\ -Q_n & -A_f^\top \end{bmatrix} = \begin{bmatrix} I_n & O_n \\ X_f & I_n \end{bmatrix} \begin{bmatrix} \hat{A}_f & -G_f \\ O_n & -\hat{A}_f^\top \end{bmatrix} \begin{bmatrix} I_n & O_n \\ X_f & I_n \end{bmatrix}^{-1} ,$$

where $\hat{A}_f = A_f - B_f K_f = A_f - G_f X_f$ is the closed-loop matrix in the faulty case, with I_n and O_n the identity and null matrices, respectively.

Matrix-Sign-Function approach IV

Henceforth, introduce the matrix

$$J = \begin{bmatrix} O_n & I_n \\ -I_n & O_n \end{bmatrix} \in \mathbb{R}^{2n \times 2n},$$

and observe that H_f is J-symmetric, i.e., $JH_f = (JH_f)^\top$; that is, if $\lambda_f \in \Lambda(H_f)$, then $-\lambda_f \in \Lambda(H_f)$.

Hence, the spectrum of the Hamiltonian-matrix is symmetric w.r.t. the imaginary axis and therefore H_f has exactly n stable eigenvalues, if there are no eigenvalues with $\mathfrak{Re}\,(\lambda_f) = 0$.

Remember also that there always exists a similarity transformation T_f of eigenvectors such that

$$H_f = T_f \begin{bmatrix} H_f^- & O_n \\ O_n & H_f^+ \end{bmatrix} T_f^{-1}, \tag{7}$$

where H_f^- and H_f^+ are Jordan blocks associated with the stable and unstable eigenvalues, respectively.

Matrix-Sign-Function approach V

Then, following the property of matrix functions saying that, for any nonsingular T_f, one has

$$f(T_f A_f T_f^{-1}) = T_f f(A_f) T_f^{-1} ,$$

see that $\text{sign}\,(H_f)$ may be computed upon (7) as

$$\text{sign}\,(H_f) = T_f \begin{bmatrix} -I_n & O_n \\ O_n & I_n \end{bmatrix} T_f^{-1} . \tag{8}$$

Moreover, the Newton iteration for computing the principal square-root of the identity matrix

$$Z_{f_{k+1}} = \frac{1}{2}\left(Z_{f_k} + Z_{f_k}^{-1}\right) , \text{ for } k = 1, 2, \ldots, N_{MSF} , \tag{9}$$

with the initial step $Z_{f_0} = H_f$, obeys

$$\lim_{k \to +\infty} Z_{f_k} = \text{sign}\,(H_f) = Z_{f_{N_{MSF}}} .$$

Matrix-Sign-Function approach VI

So forth, the convergence point may be expressed as

$$\begin{bmatrix} Z_{f_{11}} & Z_{f_{12}} \\ Z_{f_{21}} & Z_{f_{22}} \end{bmatrix} = \begin{bmatrix} I_n & O_n \\ X_f & I_n \end{bmatrix} \begin{bmatrix} -I_n & W_f \\ O_n & I_n \end{bmatrix} \begin{bmatrix} I_n & O_n \\ X_f & I_n \end{bmatrix}^{-1},$$

where W_f is the solution to the continuous algebraic Lyapunov equation

$$\hat{A}_f W_f + W_f \hat{A}_f^\top - 2G_f = 0\,.$$

Hence, the solution $X_{f_{N_{MSF}}}$ to (10), respectively

$$CARE_f: \quad A_f^\top X_f + X_f A_f - X_f B_f R_n^{-1} B_f^\top X_f + Q_n = 0\,, \tag{10}$$

results by solving the over-determined but consistent system

$$\begin{bmatrix} Z_{f_{k_{12}}} \\ Z_{f_{k_{22}}} + I_n \end{bmatrix} X_{f_k} = -\begin{bmatrix} Z_{f_{k_{11}}} + I_n \\ Z_{f_{k_{21}}} \end{bmatrix}, \tag{11}$$

for $k =_{k\to+\infty} N_{MSF}$, after partitioning $Z_{f_{N_{MSF}}}$ accordingly.

Symmetry of Matrix-Sign-Function approach I

Still, accelerating the speed of the first steps is possible through an appropriate scaling and by symmetrizing the iterations, i.e., replacing (9) with

$$\bar{Z}_{f_k} = Z_{f_k} \left[\det(Z_{f_k})\right]^{-\frac{1}{2n}} \text{ and} \tag{12a}$$

$$Z_{f_{k+1}} = \bar{Z}_{f_k} - \frac{1}{2}\left[\bar{Z}_{f_k} - (J\bar{Z}_{f_k})^{-1} J\right] . \tag{12b}$$

Another improvement might be achieved if avoiding the inversion in iterations $(12b)$ by using the Newton-Schulz scaling

$$Z_{f_{k+1}} = \frac{1}{2} Z_{f_k} \left(3I_{2n} - Z_{f_k}^2\right) , \text{ for } Z_{f_0} = H_f ; \tag{13}$$

though (13) involves no matrix inverses, unfortunately it highly depends on the norm of the post-fault Hamiltonian-matrix that might be far from 1, whereas the convergence of (13) is guaranteed only for $\|I_{2n} - H_f\|_2 < 1$.

Also, instead of the determinantal scaling in (12a), the less numerically expensive Frobenius norm or Barraud scaling might be used; scaling improves consistently the numerical conditioning of the method. The Matrix-Sign-Function method can also be applied either to solving generalized Riccati and Lyapunov equations, or to computing invariant-subspaces.

Convergence of Matrix-Sign-Function approach I

This far, the computational load required by the Matrix-Sign-Function to obtain the solution to the post-fault CARE is averaged at $4\,n^3$ per iteration. The complexity of this approach might be considerably raised whether solving at each step the system of equations (11), solution of which is better numerically conditioned if obtained using the QR-algorithm instead of the pseudo-inverse matrix.

Concluding, the Matrix-Sign-Function method converges to the solution of CARE$_f$ no matter if the post-fault system becomes unstable after the fault by simply initializing the algorithm with $Z_{f_0} = H_f$.

On the contrary, for the Newton-Kleinman approach, this technique obviously can be initialized using the solution to CARE$_n$, i.e., using X_n which gave the nominal control matrix K_n until the moment the fault occurred only if the system remains stable after a fault.

Nonetheless, in situations when the closed-loop system matrix $\hat{A}_f = A_f - B_f K_n$ eventually becomes unstable, an appropriate stabilizing initial step is crucial, which is necessary to ensure the convergence of the method. This aspect is discussed in the sequel.

Monotonicity of Matrix-Sign-Function approach I

In the case the minimum singular value of the Hamiltonian-matrix H_f is very small, a similar version of Bass-Armstrong spectrum-shifting procedure can also be applied to the Matrix-Sign-Function but on H_f's spectrum this time.

This technique amounts to starting the sign-iterations in (9) with the initial step

$$Z_{f_0} = H_f - \lambda_{\text{avg}} \times I_{2n} \,,$$

with

$$\lambda_{\text{avg}} = \frac{1}{2n} \text{tr}\left(H_f\right) ,$$

where $\text{tr}\left(.\right)$ stands for the trace of the square matrix in the argument.

That might be motivated by an unbalanced nominal matrix A_n and heretofore the post-fault one A_f which could contain elements with different orders of magnitude. Thus, an appropriate balancing to bring equilibrium between the numerical values that express physical quantities with different metrics should be done before beginning the iteration.

Monotonicity of Matrix-Sign-Function approach II

In any event, the difficulty in applying the Matrix-Sign-Function method for progressive accommodation is that no "anytime" result can be established for its convergence and hence it is impossible to see how the cost of the control solution evolves along the iterations; that is, against choosing the progressive accommodation approach based on the Matrix-Sign-Function method stands the fact that nothing can be said about the ordering of the series of intermediate solutions, as the definiteness of X_{f_k}'s obtained at each iteration fluctuates following no pattern (this is eventually to be seen in the simulations, but it was also questioned before in the literature).

Cuprins

1. Matrix Sign Function
2. B747 Application Example
   - Iterative Fault Accommodation - Matrix-Sign-Function Approach
   - Faulty System - Unstable Case

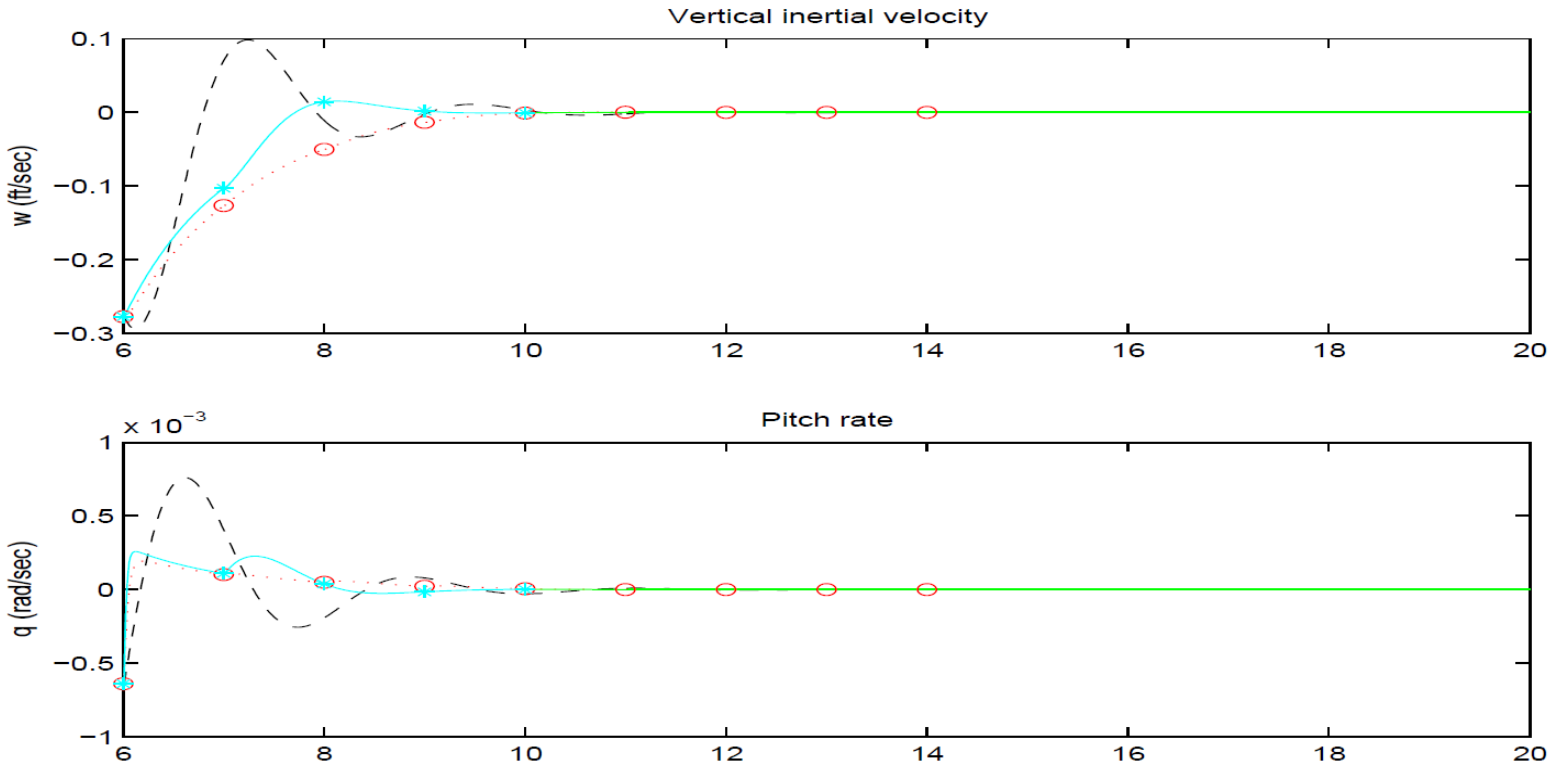
Iterative fault accommodation - Matrix-Sign-Function approach I

Furthermore, the following table shows the complete accommodation of the control matrix K_f^{acc} along the $N_{MSF} = 4$ iterations.

k	1	2	3	4
$k_{f_1}^{acc}$	-0.3988	-0.0895	-0.0812	-0.0812
$k_{f_2}^{acc}$	-386.5720	-19.9963	-26.8122	-26.8122

Iterative fault accommodation - Matrix-Sign-Function approach II

The following figure shows how the accommodation process using Matrix-Sign-Function is faster in number of steps but still Newton-Kleinman is smoother again.



NK- / MSF- PAS vs. CAS for B747-SP - Stable Case (Transient)

Faulty System - Unstable Case I

Recall that there is no need for a special initialization procedure in this case; using the same criterion as previously for testing convergence, the method needed again only $N_{MSF} = 4$ steps for the accommodation of the post-fault closed-loop system. The evolution of the control gains in the Matrix-Sign-Function approach is shown in the following table.

k	1	2	3	4
$k_{f_1}^{acc}$	-4.3076	-1.1391	-2.4510	-2.4510
$k_{f_2}^{acc}$	-395.7656	-71.1889	-216.9582	-216.9582

Questions? vs. Answers!

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